

$$B = (B^T B)^T = [Y^T X (X^T X)^{-1}]^T$$

$$= ((X^T X)^{-1})^T (Y^T X)^T$$

$$= (X^T X)^{-1} (X^T Y)$$

(X)
so good
of T

$$B = (X^T X)^{-1} X^T Y$$

(n+1) x 1

for $n=999$.

$O(n^3)$

time

complexity

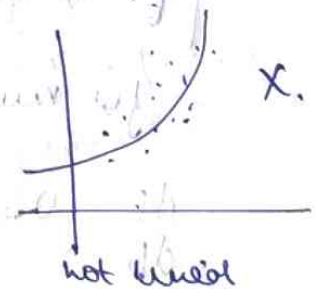
$$1000^3 = 10^9 \text{ time}$$

(Slow)

(hence gradient descent used)

Assumptions of LR:

1. Linear relationship b/w data



Relationship b/w

IP and O/P columns

should be linear.

No

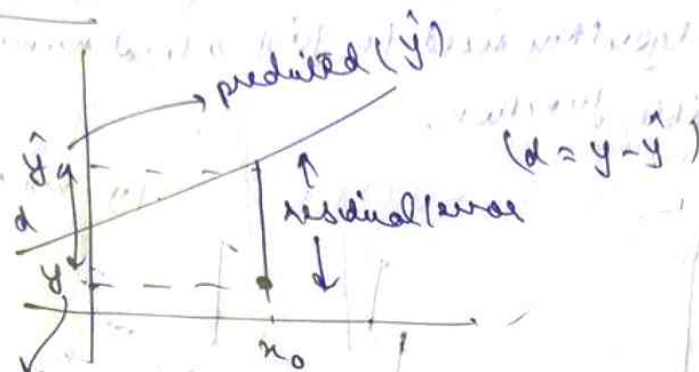
2. Multi collinearity: There should be no multicollinearity (relation b/w IP cols)

$X_1 \quad X_2 \quad X_3 \quad | \quad Y$

No correlation

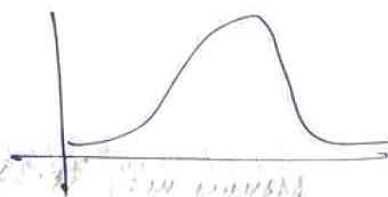
(should be independent)

3. Normal Residual



$$\text{residual} = (y_{\text{test}} - y_{\text{pred}})$$

[The residual should be normally distributed]



4. ~~homoscedasticity~~ homoscedasticity: "Having the same scatter"

Spread of residual should be same/equal.

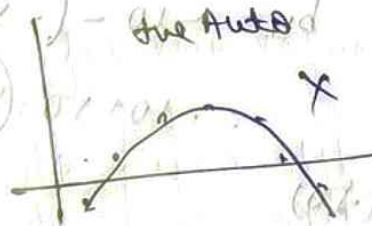
Draw scatterplot (x-axis $\rightarrow y_{\text{pred}}$, y-axis $\rightarrow$ residual)

Spread should be uniform.

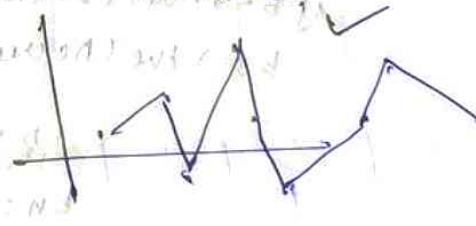
5. Autocorrelation of residuals: there should be no pattern b/w.

residuals

(01) - the Auto



(the auto) - the Auto



plt.plot(residual)